Advanced Steganalysis & Payload Processing

A Multi-Layered Threat Detection System Utilizing AVL Trees

Project Presentation

1. Introduction to Steganography

- What is Steganography?
  - The practice of concealing a file, message, image, or video within another file.
  - Unlike cryptography, which hides meaning, steganography hides existence.
  - Common technique: Least Significant Bit (LSB) Replacement.
- The Challenge:
  - Hidden payloads can bypass standard security filters.
  - Advanced embedding techniques evade simple visual inspection.

2. Problem Statement

- Limitations of Conventional Approaches:
  - Pure CNN Models: SRNet is powerful but can fail to detect non-standard LSB payloads.
  - Single-Method Extractors: Relying on one LSB extraction path misses hidden data in specific color channels.
- Our Goal:
  - Develop a robust, multi-layered Steganalysis application combining deep learning, statistical anomaly detection, multi-method extraction, and AVL Trees.

3. System Architecture

- A decoupled, modern architecture designed for speed and reliability.
- Frontend: Responsive, premium UI (HTML/CSS/JS) for image upload and reporting.
- Backend: High-performance FastAPI server.
- Machine Learning Core: PyTorch-based SRNet (Steganalysis Residual Network).
- Extraction Engine: Custom Python modules for statistical and byte-level LSB analysis.
- Data Structure: AVL Tree for deterministic, sorted payload character storage.

4. Detection Pipeline - Stage 1: Deep Learning

- SRNet (Steganalysis Residual Network):
  - A Convolutional Neural Network specifically designed for steganalysis.
  - Input: 256x256 normalized image tensor.
  - Process: Learns noise residuals rather than image content.
  - Output: A probability score indicating the likelihood of stego-image.
- Limitation: Might miss very low payload capacities; requiring further analysis.

5. Detection Pipeline - Stage 2: Extraction

- To ensure no hidden text is missed, we aggressively extract bits:
- 1. Stegano Library Extract: Standard LSB decoding.
- 2. Raw RGB LSB: Reads the LSB of R, G, and B channels for every pixel.
- 3. Raw RED LSB: Reads only the LSB of the Red channel.
- 4. Sequential Plane Extract: Reads the entire Red plane, then Green, then Blue.
- Bits are aggregated and translated into 8-bit ASCII until a null-terminator is hit.

6. Detection Pipeline - Stage 3: Statistical Analysis

- Sample-Pair Analysis (Chi-Square Test):
  - Concept: In a clean image, adjacent pixel-value pairs (2k, 2k+1) histogram is uneven.
  - LSB Effect: Random bit flipping forces adjacent pairs toward equality.
  - Detection: We calculate the normalised Chi-square value.
  - Threshold: A normalised score < 1.5 is highly suspicious.
  - Result: Flags the image even if no readable ASCII payload was extracted.

7. The Core Data Structure: AVL Trees

- Processing extracted messages using an AVL Tree:
- Why an AVL Tree?
  - Self-balancing Binary Search Tree where height difference is at most 1.
  - Guarantees O(log n) time complexity for insertions.
- Our Implementation:
  - Each node stores an extracted character.
  - Handles balancing via Left-Right (LR) and Right-Right (RR) rotations.
  - Inorder Traversal retrieves payload characters in a sorted, deterministic order.

8. Decision Fusion Engine

- Aggregating all stages for a final determination:
  - Definite Threat: If ANY extraction method finds a readable payload.
  - Likely Threat: No readable message, but Chi-Square indicates statistical anomaly (< 1.5).
  - Potential Threat: CNN detects anomalies but no payload or statistical flags triggered.
- This multi-layered net guarantees minimal false negatives.

9. Application UI & Reporting

- The forensic report includes:
- 1. Status: Clean or Stego Image.
- 2. Confidence Score: 0% to 100%.
- 3. Hidden Message: The exact extracted ASCII payload (or anomaly warning).
- 4. AVL Tree Inorder: The sorted characters of the payload.
- 5. Chi-Square Score: The raw statistical variance metric.

10. Conclusion & Future Scope

- Conclusion:
  - Combining SRNet, deterministic LSB extraction, Sample-Pair Analysis, and AVL trees creates a highly resilient platform.
- Future Scope:
  - Add support for spatial domain steganography beyond LSB (e.g., PVD).
  - Support for frequency-domain detection (JPEG DCT).
  - Visualize the AVL Tree directly in the Frontend UI.